

JOAQUIN ARREDONDO

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Education

California Polytechnic State University, San Luis Obispo **Sep 2022 – Jun 2026**
B.S. Computer Science; Minor in Computing for Interactive Arts; GPA: 3.7
Coursework: Operating Systems, Systems Programming, Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Distributed Computing, Real-Time Computer Graphics, Mixed Reality, Game Design, 3D Modeling, Digital Art, Illustration.

Projects

Path of Anu – VR Spellcasting Puzzle Game | *Unity, C#, XR Toolkit, Houdini* **Jan – Jun 2026**

- Architected C# game managers, reusable interactables, and event-driven communication for interconnected spellcasting puzzles.
- Programmed spell input to drive puzzle state changes, environmental reactions, VFX, SFX, animation, and progression with responsive feedback.
- Integrated interactive audio, music, lighting, post-processing, particle effects, and 3D assets into a cohesive VR experience.
- Built procedural Houdini Digital Assets and refined narrative, level flow, and usability through storyboards, client reviews, prototyping, and playtesting.

Seasick Scallywags – Naval Dungeon Crawler | *Unity, C#, Git* **Mar – Jun 2026**

- Led gameplay and visual development for two large levels, including terrain, lighting, post-processing, navigation cues, and encounter progression.
- Engineered chunk management, enemy lifecycle, and scalable spawning systems that culled inactive geometry and entities to improve frame rate.
- Implemented six enemy archetypes and boss behaviors with custom AI, combat, projectiles, animation controllers, VFX, SFX, and progressive challenge.
- Collaborated through Git, resolved Unity merge conflicts, mentored three developers, and used playtests to redesign terrain, guidance, and pacing.

VR Ocean Site One – Biodiversity Experience | *Unity, C#, XR Toolkit, Maya, Substance Painter* **Sep 2025 – Jan 2026**

- Modeled, textured, rigged, weight-painted, and animated shark and barracuda assets using IK systems and deformers for a VR mini-game.
- Integrated assets into Unity and developed animation controllers, state machines, animation events, and C# scripts for responsive predator AI.
- Built reusable audio managers and a mixer, documented the Maya-to-Unity pipeline, and iterated through Agile client reviews and user testing.

CYMKingdom – Real-Time Graphics Engine | *C++, OpenGL, GLSL, Unity, Photoshop* **Mar – Jun 2025**

- Authored a three-biome world with Unity Terrain Tools and Photoshop, exporting heightmaps, splatmaps, and texture masks for engine integration.
- Implemented C++ and GLSL terrain reconstruction, material blending, chunk-managed GPU instancing, and grass placement masks for efficient rendering.
- Developed Blinn-Phong lighting, shadow mapping, model and material loading, camera controls, and MVP transformations in OpenGL.

Experience

California Polytechnic State University **Jun 2025 – Present**
Undergraduate Researcher – Emotionally Responsive NPC Systems *San Luis Obispo, CA*

- Scripted a modular Utility AI brain that scored NPC needs, emotions, personality, relationships, and available actions to select behaviors.
- Designed data-driven interaction and decision frameworks with Scriptable Objects, weighted considerations, MBTI traits, and PAD emotional states.
- Connected AI decisions to autonomous navigation and destination selection, then built a low-poly sandbox to test interactions and emergent behavior.

California Polytechnic State University **Sep 2025 – Jun 2026**
Instructional Student Assistant – Game Design and Interactive Entertainment *San Luis Obispo, CA*

- Mentored 30+ students across two courses and reviewed Unity, C#, and JavaScript projects for gameplay architecture, usability, visual feedback, and code quality.

Leadership & Activities

Sigma Phi Delta Professional Engineering Fraternity **2023 – 2026**
Executive Board Member; Justice, Equity, Diversity & Inclusion (JEDI) Chair

- Led justice, equity, diversity, inclusion, recruitment, mentorship, and engagement initiatives for a 60-member chapter.

Polyphonics, California Polytechnic State University **2022 – 2023**
Auditioned Ensemble Singer

- Performed throughout California in a 60-member auditioned ensemble, strengthening musicianship, teamwork, and communication.

Technical Skills

Programming: Unity, C#, C++, C, Python, Java, JavaScript, OpenGL, GLSL, Git

Art Tools: Autodesk Maya, Houdini, Substance Painter, Adobe Photoshop, Illustrator, InDesign, Procreate